

Execution Environment

Author: u64322043
File: /home/u64322043/Personal Project 3 - Average Price Based on the Factors in India/Personal Project 3 SAS Code.sas
SAS Platform: Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host: ODAWS01-USW2-2.ODA.SAS.COM
SAS Version: 9.04.01M8P02222023
SAS Locale: en_US
Submission Time: 3/17/2026, 11:17:58 PM
Browser Host: 99.209.173.182
User Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36
Application Server: ODAMID00-USW2-2.ODA.SAS.COM

Code: Personal Project 3 SAS Code.sas

```
/* Sicheng Lu */
* Business Analytics Post Grad;
* Bachelor of Economics Honours with Minor in Statistics;
* Personal Project 3: Flight Prices based on the factors in India;
* SAS Regression Modelling;

%web_drop_table(WORK.IMPORT);
/* Importing the data file to this SAS */

FILENAME REFFILE '/home/u64322043/Personal Project 3 - Average Price Based on the Factors in India/Clean_Dataset.csv';

PROC IMPORT DATAFILE=REFFILE
    DBMS=CSV
    OUT=flight_prices_india;
    GETNAMES=YES;
RUN;
* Double check the data has been uploaded successfully;
PROC CONTENTS DATA=flight_prices_india; RUN;
/* Printing the table for the first 10 datasets */
proc print data=flight_prices_india (obs=10);
title 'The first 10 datasets';
* In order to do modelling regression,
all category variable will convert to numerical variable;
data updated_data;
set flight_prices_india;
if airline = 'Vistara' then airline_num = 1;
if airline = 'Air India' then airline_num = 2;
if airline = 'SpiceJet' then airline_num = 3;
if airline = 'GO_FIRST' then airline_num = 4;
if airline = 'Indigo' then airline_num = 5;
if airline = 'AirAsia' then airline_num = 6;

if stops = 'zero' then num_stops = 0;
if stops = 'one' then num_stops = 1;
if stops = 'two' then num_stops = 2;

if destination_city = 'Bangalore' then dest = 1;
if destination_city = 'Chennai' then dest = 2;
if destination_city = 'Delhi' then dest = 3;
if destination_city = 'Hyderabad' then dest = 4;
if destination_city = 'Kolkata' then dest = 5;
if destination_city = 'Mumbai' then dest = 6;

if class = 'Busines' then classes = 1;
if class = 'Economy' then classes = 2;

run;
* Regression Modelling;
proc reg data = updated_data;
```

```
model price = airline_num num_stops dest classes days_left;
title 'Regression Model for Average Price Based on the Factors';
run;

* SAS Graph;

proc sgplot data=updated_data;
  reg y=price x=days_left / group = classes;
  title "Flight Price vs Days Left by Class";
run;

* class = 1 = Business = Red
* class = 2 = Economy = Blue
```

Log: Personal Project 3 SAS Code.sas

Warnings (1)

Notes (15)

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69      /* Sicheng Lu */
70      * Business Analytics Post Grad;
71      * Bachelor of Economics Honours with Minor in Statistics;
72      * Personal Project 3: Flight Prices based on the factors in India;
73      * SAS Regression Modelling;
74
75      %web_drop_table(WORK.IMPORT);
76      /* Importing the data file to this SAS */
77
78      FILENAME REFFILE '/home/u64322043/Personal Project 3 - Average Price Based on the Factors in India/Clean_Dataset.csv';
79
80      PROC IMPORT DATAFILE=REFFILE
81      DBMS=CSV
82      OUT=flight_prices_india;
83      GETNAMES=YES;
84      RUN;
```

NOTE: Import cancelled. Output dataset WORK.FLIGHT_PRICES_INDIA already exists. Specify REPLACE option to overwrite it.
NOTE: The SAS System stopped processing this step because of errors.
NOTE: PROCEDURE IMPORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	305.40k
OS Memory	29348.00k
Timestamp	03/18/2026 03:17:57 AM
Step Count	370 Switch Count 0
Page Faults	0
Page Reclaims	14
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	0

```
85      * Double check the data has been uploaded successfully;
```

```
86      PROC CONTENTS DATA=flight_prices_india; RUN;
```

NOTE: PROCEDURE CONTENTS used (Total process time):

real time	0.02 seconds
user cpu time	0.03 seconds
system cpu time	0.00 seconds
memory	3365.50k
OS Memory	30636.00k
Timestamp	03/18/2026 03:17:57 AM
Step Count	371 Switch Count 0
Page Faults	0
Page Reclaims	294
Page Swaps	0
Voluntary Context Switches	3
Involuntary Context Switches	3

```
Block Input Operations      0
Block Output Operations     16
```

```
87      /* Printing the table for the first 10 datasets */
88      proc print data=flight_prices_india (obs=10);
89      title 'The first 10 datasets';
90      * In order to do modelling regression,
91      all category variable will convert to numerical variable;
```

NOTE: There were 10 observations read from the data set WORK.FLIGHT_PRICES_INDIA.

NOTE: PROCEDURE PRINT used (Total process time):

```
real time      0.02 seconds
user cpu time   0.02 seconds
system cpu time 0.00 seconds
memory         2394.50k
OS Memory      30376.00k
Timestamp      03/18/2026 03:17:57 AM
Step Count     372  Switch Count  1
Page Faults    0
Page Reclaims  253
Page Swaps     0
Voluntary Context Switches 10
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 16
```

```
92      data updated_data;
93      set flight_prices_india;
94      if airline = 'Vistara' then airline_num = 1;
95      if airline = 'Air India' then airline_num = 2;
96      if airline = 'SpiceJet' then airline_num = 3;
97      if airline = 'GO_FIRST' then airline_num = 4;
98      if airline = 'Indigo' then airline_num = 5;
99      if airline = 'AirAsia' then airline_num = 6;
100
101      if stops = 'zero' then num_stops = 0;
102      if stops = 'one' then num_stops = 1;
103      if stops = 'two' then num_stops = 2;
104
105      if destination_city = 'Bangalore' then dest = 1;
106      if destination_city = 'Chennai' then dest = 2;
107      if destination_city = 'Delhi' then dest = 3;
108      if destination_city = 'Hyderabad' then dest = 4;
109      if destination_city = 'Kolkata' then dest = 5;
110      if destination_city = 'Mumbai' then dest = 6;
111
112      if class = 'Busines' then classes = 1;
113      if class = 'Economy' then classes = 2;
114
115      run;
```

NOTE: There were 300153 observations read from the data set WORK.FLIGHT_PRICES_INDIA.

NOTE: The data set WORK.UPDATED_DATA has 300153 observations and 16 variables.

NOTE: DATA statement used (Total process time):

```
real time      0.06 seconds
user cpu time   0.04 seconds
system cpu time 0.02 seconds
memory         3576.09k
OS Memory      32428.00k
Timestamp      03/18/2026 03:17:57 AM
Step Count     373  Switch Count  6
Page Faults    0
Page Reclaims  528
Page Swaps     0
Voluntary Context Switches 27
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 75272
```

```
116      * Regression Modelling;
117      proc reg data = updated_data;
118      model price = airline_num num_stops dest classes days_left;
119      title 'Regression Model for Average Price Based on the Factors';
120      run;
```

WARNING: ODS graphics with more than 5000 points have been suppressed. Use the PLOTS(MAXPOINTS=) option in the PROC REG statement to change or override the cutoff.

```
121
122      * SAS Graph;
```

NOTE: PROCEDURE REG used (Total process time):

real time	0.10 seconds
user cpu time	0.07 seconds
system cpu time	0.04 seconds
memory	5157.56k
OS Memory	33984.00k
Timestamp	03/18/2026 03:17:57 AM
Step Count	374
Switch Count	7
Page Faults	0
Page Reclaims	742
Page Swaps	0
Voluntary Context Switches	59
Involuntary Context Switches	3
Block Input Operations	0
Block Output Operations	110144

```
123      proc sgplot data=updated_data;
124          reg y=price x=days_left / group = classes;
125          title "Flight Price vs Days Left by Class";
126      run;
```

NOTE: PROCEDURE SGPLOT used (Total process time):

real time	1.20 seconds
user cpu time	0.78 seconds
system cpu time	0.07 seconds
memory	23401.40k
OS Memory	50072.00k
Timestamp	03/18/2026 03:17:58 AM
Step Count	375
Switch Count	41
Page Faults	0
Page Reclaims	10210
Page Swaps	0
Voluntary Context Switches	347
Involuntary Context Switches	11
Block Input Operations	0
Block Output Operations	86152

NOTE: Marker and line antialiasing has been disabled for at least one plot because the threshold has been reached. You can set ANTIALIASMAX=300200 in the ODS GRAPHICS statement to enable antialiasing for all plots.

NOTE: There were 300153 observations read from the data set WORK.UPDATED_DATA.

```
127      * class = 1 = Busines = Red
128      * class = 2 = Economy = Blue
129
130
131
132      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
133      ODS HTML CLOSE;
134      &GRAPHTERM; ;*';*";*/;RUN;QUIT;
135      QUIT;RUN;
136      ODS HTML5 (ID=WEB) CLOSE;
137
138      FILENAME _GSFNAME;
NOTE: Fileref _GSFNAME has been deassigned.
139      DATA _NULL_;
140      RUN;
```

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	471.34k
OS Memory	29100.00k
Timestamp	03/18/2026 03:17:58 AM
Step Count	376
Switch Count	0
Page Faults	0
Page Reclaims	23
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	0

Results: Personal Project 3 SAS Code.sas

The CONTENTS Procedure

Data Set Name	WORK.FLIGHT_PRICES_INDIA	Observations	300153
Member Type	DATA	Variables	12
Engine	V9	Indexes	0
Created	03/17/2026 22:34:23	Observation Length	96
Last Modified	03/17/2026 22:34:23	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	221
First Data Page	1
Max Obs per Page	1363
Obs in First Data Page	1327
Number of Data Set Repairs	0
Filename	/saswork/SAS_work8AEC0000C4AB_odaws02-usw2-2.oda.sas.com/SAS_work26A40000C4AB_odaws02-usw2-2.oda.sas.com/flight_prices_india.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1946180355
Access Permission	rW-r--r--
Owner Name	u64322043
File Size	28MB
File Size (bytes)	29097984

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Informat
1	VAR1	Num	8	BEST12.	BEST32.
2	airline	Char	9	\$9.	\$9.
7	arrival_time	Char	13	\$13.	\$13.
9	class	Char	7	\$7.	\$7.
11	days_left	Num	8	BEST12.	BEST32.
5	departure_time	Char	13	\$13.	\$13.
8	destination_city	Char	6	\$6.	\$6.
10	duration	Num	8	BEST12.	BEST32.
3	flight	Char	7	\$7.	\$7.
12	price	Num	8	BEST12.	BEST32.
4	source_city	Char	5	\$5.	\$5.
6	stops	Char	4	\$4.	\$4.

The first 10 datasets

Obs	VAR1	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	duration	days_left	price
1	0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	2.17	1	5953
2	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.33	1	5953
3	2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	2.17	1	5956
4	3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.25	1	5955
5	4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	2.33	1	5955
6	5	Vistara	UK-945	Delhi	Morning	zero	Afternoon	Mumbai	Economy	2.33	1	5955
7	6	Vistara	UK-927	Delhi	Morning	zero	Morning	Mumbai	Economy	2.08	1	6060
8	7	Vistara	UK-951	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.17	1	6060
9	8	GO_FIRST	G8-334	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	2.17	1	5954
10	9	GO_FIRST	G8-336	Delhi	Afternoon	zero	Evening	Mumbai	Economy	2.25	1	5954

Regression Model for Average Price Based on the Factors

The REG Procedure
Model: MODEL1
Dependent Variable: price

Number of Observations Read	300153
Number of Observations Used	82405
Number of Observations with Missing Values	217748

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	3.874667E13	7.749335E12	150006	<.0001
Error	82399	4.256752E12	51660241		
Corrected Total	82404	4.300343E13			

Root MSE	7187.50588	R-Square	0.9010
Dependent Mean	19022	Adj R-Sq	0.9010
Coeff Var	37.78500		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	96493	148.11895	651.46	<.0001
airline_num	1	-355.49930	15.39379	-23.09	<.0001
num_stops	1	8192.39549	65.79045	124.52	<.0001
dest	1	404.70180	16.74447	24.17	<.0001
classes	1	-47000	65.71229	-715.23	<.0001
days_left	1	-138.64953	1.85673	-74.67	<.0001

